Razeen Khan

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EDUCATION

University of California, San Diego — Expected: June 2028

Bachelor of Science (B.S.) Mathematics-Computer Science

Relevant Coursework: Advanced Data Structures, Object-Oriented Design, Systems Programming, Graph Theory

EXPERIENCE

Software Engineering Intern — July 2025 - October 2025

General Atomics & San Diego Supercomputer Center — *San Diego, CA*

- Reduced experimental data retrieval latency by **40%** by optimizing **SQL** queries and relational database indexing
- Implemented Python data pipelines to ingest, validate, and preprocess large experimental datasets with NumPy
- Automated **Linux** cluster workflows using Bash, SLURM, and SSH, reducing manual analysis time by **60%**
- Developed reusable backend modules processing AWS S3 and internal API data, improving efficiency by 12%

Software Engineering Intern — April 2025 - Present

UC San Diego School of Medicine — *San Diego, CA*

- Developed production Python modules on the GenePattern genomic analysis platform serving **100,000+ users**
- Improved analytics throughput by **55%** via **Docker** containerization, parallel execution and CI/CD pipelines
- Reduced runtime failures by **40%** using Kubernetes, structured logging, validation, and exception handling
- Authored internal and external docs, shortening developer onboarding by 30% and improving user experience

Data Engineer — March 2025 - Present

Student Foundation Investment Committee, UCSD — *San Diego, CA*

- Processed and aligned **50+ years** of macro indicators and market returns using pandas, NumPy, and matplotlib
- Built a backend service exposing analytics and model outputs via **RESTful APIs** with authentication and caching
- Reduced data processing and backtesting runtime by **65%** through vectorized preprocessing and caching
- Developed ML classification models via **PyTorch** and scikit-learn, improving prediction accuracy by **30%**

Product Manager — April 2025 - August 2025

Youphoria — *Remote*

- Defined technical requirements for backend services and APIs, improving development efficiency by **20%**
- Increased feature adoption by **20%** by prioritizing features using usage metrics and system constraints on Swift

PROJECTS

Claimi: AI Agent for Automated Settlement Claims *(NexHacks 2026)*

- Built multi-agent AI system using **MCP** for web navigation, data extraction, and automated form submission
- Designed **planner–executor architecture** with retry, self-verification, and step-level checkpoints
- Implemented **RAG-style** document retrieval for eligibility rules and evidence requirements, reducing API costs
- Built **parallel task execution** and batching to process claims concurrently and validate **DOM** parsing
- Reduced claim submission time from **~45 min to <2 min** across heterogeneous settlement claim websites

Fortuna: EdTech Entrepreneurship Mobile App

- Reduced backend API latency by **35%** through caching, and refactored endpoints; integrated Google Gemini API
- Designed a scalable persistence layer using PostgreSQL and Redis for transactional data and caching
- Deployed Firebase Cloud Functions using Node.js to manage and process real-time game states for individual users
- Implemented a rate limiter and CDN functionality to improve efficiency, reducing read/write costs by **30%**
- Developed React, JavaScript and TypeScript web frontend and integrated Supabase for secure form ingestion

SKILLS

Programming Languages: Java, Python, JavaScript/TypeScript, HTML/CSS, C/C++, SQL, Dart, MATLAB

Libraries & Frameworks: Flutter, Pandas, Scikit-learn, React, Node.js, Angular, Vue.js, BeautifulSoup, JUnit, Scrum

Tools & Technologies: Git, GitHub, Docker, GCP, AWS S3, Jupyter Notebook, Linux, Unix, Conda, SSH, MCP

Databases: MySQL, PostgreSQL, Firebase, NoSQL, MongoDB, Supabase

AWARDS

NexHacks Hackathon Winner (Seda Sponsor Track, 2nd Place) — 2026

Jane Street Estimathon (UC San Diego) Winner — 2025